

In 2022, the U.S. National Oceanic and Atmospheric Administration published the average monthly temperatures in the U.S. Census report. The table shows the data for the average recorded temperatures in Jacksonville, where 1 is January.

Month	1	2	3	4	5	6	7	8	9	10	11	12
Highest Temp ($^{\circ}\text{F}$)	56	67	76	83	88	91	92	91	88	83	76	67

1. Write a quadratic equation, in vertex form, that models the temperature, y , in terms of months, x .
2. Write the quadratic equation in standard form.
3. Use the quadratic equation to determine during what months the average temperature may be 80°F .

William is playing golf when he hits a shot off the tee. The height of the ball, $h(x)$, in feet, is modeled in the table in terms of x seconds, that the ball has been in the air.

Seconds, x	1	2	3	4
Height, $h(x)$	80	128	144	128

4. Write a quadratic equation, in vertex form, that models the height of the ball in terms of seconds, x .

5. Rewrite the quadratic equation modeling William's golf ball's height in standard form.
6. Use the quadratic equation to determine at what time the ball will land back on the ground.

A mathematician for a music company tracked the sales, in tens of millions of dollars, of music cassette tapes, y , for x years since 1980 up to the year 2000. A table of values of some of the values is shown.

7. Write a quadratic equation, in vertex form, that models the sales from music cassette tapes, y , in terms of years since 1980, x .

x	y
0	83
1	134
3	218
6	299
9	326
10	323
13	278
17	134
19	26

8. Rewrite the quadratic equation in standard form.

9. Use the quadratic equation to determine when $y = 0$.

10. Interpret the solutions and determine if the solutions are viable in context.